Dr Rémi Zallot

Senior Lecturer (Associate Professor) in Biochemistry
School of Biological and Chemical Sciences, Manchester Metropolitan University, UK

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Academic Positions

Senior Lecturer (Associate Professor) in Biochemistry Aug 2024 – present
School of Biological and Chemical Sciences, Manchester Metropolitan University, UK

Lecturer (Assistant Professor) in Biochemistry Jun 2023 – Aug 2024
Department of Life Sciences, Manchester Metropolitan University, UK

Research Associate Mar 2022 – Jun 2023
Manchester Institute of Biotechnology (MIB), The University of Manchester, UK

Marie Skłodowska-Curie Individual Fellow Oct 2019 – Mar 2022
Institute of Life Science, Swansea University Medical School, UK
"deCrYPtion" project (ID 839116), decrypting *Mycobacterium* cytochrome P450 physiological functions using comparative genomics. Supervisor: Prof. Steven L. Kelly.

Enzyme Function Initiative team 2016 – 2019
Carl R. Woese Institute for Genomic Biology, University of Illinois at Urbana-Champaign, USA
Enhanced the EFI genomic enzymology web tools; characterised genes from the human gut microbiome; trained biochemists in comparative genomics approaches.

Postdoctoral Researcher 2011 – 2016
Microbiology and Cell Science Department, University of Florida, USA
Characterised genes involved in B vitamin metabolism and queuosine tRNA modification in plants and microbes.

Education

PhD, Biochemistry / Plant Biology 2008 – 2011
Laboratoire de Biogenèse Membranaire
Thesis: *Identification and characterization of a lipase expressed during Arabidopsis thaliana reserves hydrolysis*.

Master's, Plant Biology and Biotechnology 2006 – 2008
Université de Bordeaux, France

Licence, Biochemistry 2003 – 2006
Université de Bordeaux, France

Research Interests

My aim is to advance our fundamental understanding of microbes to enhance human health. I focus on characterising and understanding the functions of genes from microbes, particularly pathogens, to leverage this knowledge for developing new drugs and biotechnological applications. My approach is multidisciplinary, combining bioinformatics — including comparative genomics and genomic enzymology — with experimental techniques from microbiology, genetics, and biochemistry.

Grants and Funding

The Academy of Medical Sciences – Springboard Award (SBF009\1016) Aug 2024 – Jul 2026
£125,000
STEROSPRING: Characterising an Uncharted Lanosterol Catabolic Pathway in Non-Tuberculous Mycobacterial (NTM) Species.

MMU Research and Knowledge Exchange – Research Accelerator Grant Jan – Dec 2023
£7,000
Deciphering an Uncharted Sterol Catabolic Pathway and Its Relevance in Non-Tuberculous Mycobacterial Infection.

Manchester-Melbourne-Toronto (MMT) Research Funds Apr 2023
£6,960 – Grant P122678
Missing Links in Global Sulphur Metabolism. Instructor for Genomic Enzymology and Enzyme Function Initiative training, Melbourne, Australia.

Marie Skłodowska-Curie Individual Fellowship Oct 2019 – Mar 2022
€212,933 – European Commission, Horizon 2020, Grant 839116

Higher Education Investment and Recovery Fund Aug 2021 – Feb 2022
£33,954 – Higher Education Funding Council for Wales (HEFCW)

Inception Programme (Investissement d'Avenir) – Travel Grant Jun 2019
€3,000 – Grant ANR-16-CONV-0005
Workshop Instructor, "From Genes to Function: A Survival Guide to Web-Based Tools for Microbiologists," Institut Pasteur, Paris.

Sêr Cymru II COFUND Fellowship Jul 2019
£173,000 – Grant SU190
Offer declined due to concurrent MSCA Individual Fellowship.

Wales Marie Skłodowska-Curie Individual Fellowship Summer School May 2018
£2,000 travel grant

Allocation de recherche (PhD Research Allowance Award) Sep 2008 – Nov 2011
€90,000
French Ministry of Higher Education and Research (Ministère de l'Enseignement Supérieur et de la Recherche).

Preprints

[1] Dongxu Yuan, Lauren R. Hatfield, Hannah Ryan, Harry Williams, Mochen Dong, Josie E. Parker, Yi Jin, and Rémi Zallot. *CYP123, a sterol C4-methyl oxidase, acts with CYP51 to initiate mycobacterial lanosterol catabolism*. bioRxiv. 2026. DOI: 10.64898/2026.10.31.760369.

Publications

[2] Hai Minh Thai, Debbie M. Hunt, Yugen Miyahara, Manisha Priya, Htin L. Aung, Rémi Zallot, and Luiz Pedro S. de Carvalho. "A cloaked glutamate decarboxylase sustains the GABA shunt in Mycobacterium tuberculosis". In: *Proceedings of the National Academy of Sciences* (2026). DOI: 10.1073/pnas.2619778123.

[3] Caio Felipe Cavicchia Zamunér, Roobinidevi Ragupathy, Mateus S. Terceti, Rémi Zallot, James Redfern, Jeffrey B. Jones, Franklin Behlau, Henrique Ferreira, and Mark C. Enright. "Characterization of putatively lytic bacteriophages able to infect Xanthomonas citri subsp. citri and identification of novel putative exopolysaccharide depolymerases". In: *Microbial Genomics* (2026). DOI: 10.1099/mgen.0.001821.

[4] Valérie de Crécy-Lagard, Geoffrey Hutinet, José D. D. Cediel-Becerra, Yifeng Yuan, Rémi Zallot, Marc G. Chevrette, R. M. Madhushi N. Ratnayake, Marshall Jaroch, Samia Quaiyum, and Steven Bruner. "Biosynthesis and function of 7-deazaguanine derivatives in bacteria and phages". In: *Microbiology and Molecular Biology Reviews* (2024). DOI: 10.1128/mmbr.00199-23.

[5] Mochen Dong, Zhuoyun Chen, Yuan He, Rémi Zallot, and Yi Jin. "Bioinformatics-facilitated identification of novel bacterial sulfoglycosidases that hydrolyze 6-Sulfo-N-acetylglucosamine". In: *ACS Bio & Med Chem Au* (2024). DOI: 10.1021/acsbiomedchemau.4c00088.

[6] Guangcai Xu, Daniele Torri, Sebastian Cuesta-Hoyos, Deepanjan Panda, Luke R. L. Yates, Rémi Zallot, Kehan Bian, Dongxu Jia, Andreea I. Iorgu, Colin Levy, Sarah A. Shepherd, and Jason Micklefield. "Cryptic enzymatic assembly of peptides armed with β-lactone warheads". In: *Nature chemical biology* (2024). DOI: 10.1038/s41589-024-01657-7.

[7] Nils Oberg, Rémi Zallot, and John A. Gerlt. “EFI-EST, EFI-GNT, and EFI-CGFP: enzyme function initiative (EFI) web resource for genomic enzymology tools”. In: *Journal of molecular biology* (2023). DOI: 10.1016/j.jmb.2023.168018.

[8] Zhen Zhang, Mochen Dong, Rémi Zallot, George Michael Blackburn, Nini Wang, Chengjian Wang, Long Chen, Patrick Baumann, Zuyan Wu, Zhongfu Wang, Haiming Fan, Christian Roth, Yi Jin, and Yuan He. “Mechanistic and structural insights into the specificity and biological functions of bacterial sulfoglycosidases”. In: *ACS Catalysis* (2023). DOI: 10.1021/acscatal.2c05405.

[9] Qiang Li, Rémi Zallot, Brian S. MacTavish, Alvaro Montoya, Daniel J. Payan, You Hu, John A. Gerlt, Alexander Angerhofer, Valérie de Crécy-Lagard, and Steven D. Bruner. “Epoxyqueuosine reductase QueH in the biosynthetic pathway to tRNA queuosine is a unique metalloenzyme”. In: *Biochemistry* (2021). DOI: 10.1021/acs.biochem.1c00164.

[10] Rémi Zallot, Nils Oberg, and John A. Gerlt. “Discovery of new enzymatic functions and metabolic pathways using genomic enzymology web tools”. In: *Current opinion in biotechnology* (2021). DOI: 10.1016/j.copbio.2020.12.004.

[11] Yifeng Yuan, Rémi Zallot, Tyler L. Grove, Daniel J. Payan, Isabelle Martin-Verstraete, Sara Šepić, Seetharamsingh Balamkundu, Ramesh Neelakandan, Vinod K. Gadi, Chuan-Fa Liu, Manal A. Swairjo, Peter C. Dedon, Steven C. Almo, John A. Gerlt, and Valérie de Crécy-Lagard. “Discovery of novel bacterial queuine salvage enzymes and pathways in human pathogens”. In: *Proceedings of the National Academy of Sciences* (2019). DOI: 10.1073/pnas.1909604116.

[12] Rémi Zallot, Nils Oberg, and John A. Gerlt. “The EFI web resource for genomic enzymology tools: leveraging protein, genome, and metagenome databases to discover novel enzymes and metabolic pathways”. In: *Biochemistry* (2019). DOI: 10.1021/acs.biochem.9b00735.

[13] Michael S. Carter, Xinshuai Zhang, Hua Huang, Jason T. Bouvier, Brian San Francisco, Matthew W. Vetting, Nawar Al-Obaidi, Jeffrey B. Bonanno, Agnidipta Ghosh, Rémi Zallot, Harvey M. Andersen, Steven C. Almo, and John A. Gerlt. “Functional assignment of multiple catabolic pathways for D-apiose”. In: *Nature chemical biology* (2018). DOI: 10.1038/s41589-018-0067-7.

[14] Katherine J. Harrison, Valérie de Crécy-Lagard, and Rémi Zallot. “Gene Graphics: a genomic neighborhood data visualization web application”. In: *Bioinformatics* (2018). DOI: 10.1093/bioinformatics/btx793.

[15] Rémi Zallot, Nils O. Oberg, and John A. Gerlt. “‘Democratized’ genomic enzymology web tools for functional assignment”. In: *Current opinion in chemical biology* (2018). DOI: 10.1016/j.cbpa.2018.09.009.

[16] Rémi Zallot, Robert Ross, Wei-Hung Chen, Steven D. Bruner, Patrick A. Limbach, and Valérie de Crécy-Lagard. “Identification of a novel epoxyqueuosine reductase family by comparative genomics”. In: *ACS chemical biology* (2017). DOI: 10.1021/acschembio.6b01100.

[17] Rémi Zallot, Yifeng Yuan, and Valérie de Crécy-Lagard. “The Escherichia coli COG1738 member YhhQ is involved in 7-cyanodeazaguanine (preQ0) transport”. In: *Biomolecules* (2017). DOI: 10.3390/biom7010012.

[18] Ghulam Hasnain, Sanja Roje, Na Sa, Rémi Zallot, Michael J. Ziemak, Valérie de Crécy-Lagard, Jesse F. Gregory, and Andrew D. Hanson. “Bacterial and plant HAD enzymes catalyse a missing phosphatase step in thiamin diphosphate biosynthesis”. In: *Biochemical Journal* (2016). DOI: 10.1042/bj20150805.

[19] Christopher S. Henry, Claudia Lerma-Ortiz, Svetlana Y. Gerdes, Jeffrey D. Mullen, Ric Colasanti, Aleksey Zhukov, Océane Frelin, Jennifer J. Thiaville, Rémi Zallot, Thomas D. Niehaus, Ghulam Hasnain, Neal Conrad, Andrew D. Hanson, and Valérie de Crécy-Lagard. “Systematic identification and analysis of frequent gene fusion events in metabolic pathways”. In: *BMC genomics* (2016). DOI: 10.1186/s12864-016-2782-3.

[20] Lili Huang, Anna Khusnutdinova, Boguslaw Nocek, Greg Brown, Xiaohui Xu, Hong Cui, Pierre Petit, Robert Flick, Rémi Zallot, Kelly Balmant, Michael J. Ziemak, John Shanklin, Valérie de Crécy-Lagard, Oliver Fiehn, Jesse F. Gregory, Andrzej Joachimiak, Alexei Savchenko, Alexander F. Yakunin, and Andrew D. Hanson. “A family of metal-dependent phosphatases implicated in metabolite damage-control”. In: *Nature Chemical Biology* (2016). DOI: 10.1038/nchembio.2108.

[21] Manaki Mimura, Rémi Zallot, Thomas D. Niehaus, Ghulam Hasnain, Satinder K. Gidda, Thuy N.D. Nguyen, Erin M. Anderson, Robert T. Mullen, Greg Brown, Alexander F. Yakunin, Valérie de Crécy-Lagard, Jesse F. Gregory, Donald R. McCarty, and Andrew D. Hanson. “Arabidopsis TH2 encodes the orphan enzyme thiamin monophosphate phosphatase”. In: *The Plant Cell* (2016). DOI: 10.1105/tpc.16.00600.

[22] Rémi Zallot, Katherine Harrison, Bryan Kolaczkowski, and Valérie de Crécy-Lagard. “Functional annotations of paralogs: a blessing and a curse”. In: *Life* (2016). DOI: 10.3390/life6030039.

[23] Ekaterina Kuznetsova, Boguslaw Nocek, Greg Brown, Kira S. Makarova, Robert Flick, Yuri I. Wolf, Anna Khusnutdinova, Elena Evdokimova, Ke Jin, Kemin Tan, Andrew D. Hanson, Ghulam Hasnain, Rémi Zallot, Valérie de Crécy-Lagard, Mohan Babu, Alexei Savchenko, Andrzej Joachimiak, Aled M. Edwards, Eugene V. Koonin, and Alexander F. Yakunin. "Functional diversity of haloacid dehalogenase superfamily phosphatases from Saccharomyces cerevisiae: biochemical, structural, and evolutionary insights". In: *Journal of Biological Chemistry* (2015). DOI: 10.1074/jbc.m115.657916.

[24] Samuel M. D. Seaver, Svetlana Gerdes, Océane Frelin, Claudia Lerma-Ortiz, Louis M. T. Bradbury, Rémi Zallot, Ghulam Hasnain, Thomas D. Niehaus, Basma El Yacoubi, Shiran Pasternak, Robert Olson, Gordon Pusch, Ross Overbeek, Rick Stevens, Valérie de Crécy-Lagard, Doreen Ware, Andrew D. Hanson, and Christopher S. Henry. "High-throughput comparison, functional annotation, and metabolic modeling of plant genomes using the PlantSEED resource". In: *Proceedings of the National Academy of Sciences* (2014). DOI: 10.1073/pnas.1401329111.

[25] Rémi Zallot, Céline Brochier-Armanet, Kirk W. Gaston, Farhad Forouhar, Patrick A. Limbach, John F. Hunt, and Valérie de Crécy-Lagard. "Plant, animal, and fungal micronutrient queuosine is salvaged by members of the DUF2419 protein family". In: *ACS chemical biology* (2014). DOI: 10.1021/cb500278k.

[26] Rémi Zallot, Mohammad Yazdani, Aymeric Goyer, Michael J. Ziemak, Jiahn-Chou Guan, Donald R. McCarty, Valérie de Crécy-Lagard, Svetlana Gerdes, Timothy J. Garrett, Jordi Benach, John F. Hunt, David K. Shintani, and Andrew D. Hanson. "Salvage of the thiamin pyrimidine moiety by plant TenA proteins lacking an active-site cysteine". In: *Biochemical Journal* (2014). DOI: 10.1042/bj20140522.

[27] Yonghua Li-Beisson, Basil Shorrosh, Fred Beisson, Mats X. Andersson, Vincent Arondel, Philip D. Bates, Sébastien Baud, David Bird, Allan DeBono, Timothy P. Durrett, Rochus B. Franke, Ian A. Graham, Kenta Katayama, Amélie A. Kelly, Tony Larson, Jonathan E. Markham, Martine Miquel, Isabel Molina, Ikuo Nishida, Owen Rowland, Lacey Samuels, Katherine M. Schmid, Hajime Wada, Ruth Welti, Changcheng Xu, Rémi Zallot, and John Ohlrogge. "Acyl-lipid metabolism". In: *The Arabidopsis book/American Society of Plant Biologists* (2013). DOI: 10.1199/tab.0161.

[28] Mohammad Yazdani, Rémi Zallot, Meral Tunc-Ozdemir, Valérie de Crécy-Lagard, David K. Shintani, and Andrew D. Hanson. "Identification of the thiamin salvage enzyme thiazole kinase in Arabidopsis and maize". In: *Phytochemistry* (2013). DOI: 10.1016/j.phytochem.2013.05.017.

[29] Rémi Zallot, Gennaro Agrimi, Claudia Lerma-Ortiz, Howard J. Teresinski, Océane Frelin, Kenneth W. Ellens, Alessandra Castegna, Annamaria Russo, Valérie de Crécy-Lagard, Robert T. Mullen, Ferdinando Palmieri, and Andrew D. Hanson. "Identification of mitochondrial coenzyme a transporters from maize and Arabidopsis". In: *Plant physiology* (2013). DOI: 10.1104/pp.113.218081.

[30] Rémi Zallot. "Identification et caractérisation d'une lipase exprimée pendant l'hydrolyse des réserves chez Arabidopsis thaliana". PhD thesis. 2011. DOI: 10.70675/9dd49aa6zb8d6z4630z8574zc0b1ebd25e43.

[31] Yonghua Li-Beisson, Basil Shorrosh, Fred Beisson, Mats X. Andersson, Vincent Arondel, Philip D. Bates, Sébastien Baud, David Bird, Allan DeBono, Timothy P. Durrett, Rochus B. Franke, Ian A. Graham, Kenta Katayama, Amélie A. Kelly, Tony Larson, Jonathan E. Markham, Martine Miquel, Isabel Molina, Ikuo Nishida, Owen Rowland, Lacey Samuels, Katherine M. Schmid, Hajime Wada, Ruth Welti, Changcheng Xu, Rémi Zallot, and John Ohlrogge. "Acyl-lipid metabolism". In: *The Arabidopsis book/American Society of Plant Biologists* (2010). DOI: 10.1199/tab.0133.

[32] Fabienne Furt, Sabine König, Jean-Jacques Bessoule, Françoise Sargueil, Rémi Zallot, Thomas Stanislas, Elodie Noirot, Jeanine Lherminier, Françoise Simon-Plas, Ingo Heilmann, and Sébastien Mongrand. "Polyphosphoinositides are enriched in plant membrane rafts and form microdomains in the plasma membrane". In: *Plant physiology* (2010). DOI: 10.1104/pp.109.149823.

Teaching

Module Leader, Biomedical Cell Biology (BCB) 2023 – present
Manchester Metropolitan University, School of Biological and Chemical Sciences
Final-year optional module for students on the accredited Biomedical Science programme.

Medical Microbiology 2023 – present
Manchester Metropolitan University, School of Biological and Chemical Sciences
Teaching contributor; supervises final-year student projects.

Conferences and Talks

- **EFI web tools workshop, Singapore Enzyme Biology Week** — Workshop instructor. National Centre for Engineering Biology (NCEB), National University of Singapore, Singapore. October 2026.
- **Summer 2026 meeting of the Acid Fast Club** — Conference attended. Liverpool School of Tropical Medicine, Liverpool, UK. July 2026.
- **Department Summer research symposium Seminar** — Oral presentation. Manchester Metropolitan University, Department of Life Sciences, Manchester, UK. April 2026.
- **Invited Seminar** — Invited talk. Institute of Medical Research, Northwestern Polytechnical University, Xi'an, China. April 2026.
- **Invited Seminar** — Invited talk. Shanghai Jiao Tong University Medical School, Shanghai, China. April 2026.
- **Tuberculosis: Understanding the Disease Across Scales** — Poster. Keystone Symposium, Cape Town, South Africa. March 2026.
- **Winter Acid Fast Club meeting** — Conference attended. Acid Fast Club, London, UK. January 2026.
- **Leicester 2025 summer session of the Acid Fast Club meeting** — Oral presentation. Acid Fast Club, Leicester, UK. July 2025.
- **Poster presentation** — 7th Forum of Respiratory Tract Infections, Glasgow, UK. February 2025.
- **Invited talk for MMU ECR and PhD Conference** — Invited talk. Faculty of Science and Engineering, Manchester Metropolitan University, Manchester, UK. September 2024.
- **Invited talk** — National University of Singapore, Singapore. April 2024.
- **Invited virtual seminar for the Catalysis Innovation Consortium** — Invited talk. Catalysis Innovation Consortium, Virtual, USA. March 2024.
- **Genomic Enzymology and Enzyme Function Initiative (EFI) Training Workshop** — Workshop instructor. University of Melbourne, Melbourne, Australia. April 2023.
- **Poster presentation** — Gordon Research Conference on Enzymes, Coenzymes and Metabolic Pathways, Waterville Valley, NH, USA. July 2023.
- **Invited talk and EFI workshop** — National University of Singapore, Singapore. November 2023.
- **Invited talk** — Chinese Academy of Sciences, Institute of Microbiology, Beijing, China. December 2023.
- **Invited talk** — Northwest University, Xi'an, China. December 2023.
- **Understanding antimicrobial resistance requires characterising unknown genes** — Invited talk. Manchester Institute of Biotechnology (MIB) Fellowship days, The University of Manchester, Manchester, UK. January 2022.
- **Comparative genomics and genomic enzymology approaches for functional assignment** — Poster (selected for a short oral presentation). Manchester Institute of Biotechnology (MIB) Annual Science Day, The University of Manchester, Manchester, UK. 2022.
- **Similarity Networks and Genome Neighborhood Networks for function assignment** — Invited talk. School of Chemistry, Cardiff University, Cardiff, UK. April 2021.
- **Genomic enzymology web tools for functional assignment** — Invited talk. Institut Pasteur, Centre of Bioinformatics, Biostatistics and Integrative Biology, Paris, France. June 2019.
- **Quel and its associated activator QueJ are non-orthologous replacement of the nitrile reductase QueF in the queuosine biosynthesis pathway** — Poster. Gordon Research Seminar and Conference, Enzymes, Coenzymes and Metabolic Pathways, Waterville Valley, NH, USA. 2018.
- **Quel and its associated activator QueJ are non-orthologous replacement of the nitrile reductase QueF in the queuosine biosynthesis pathway** — Poster. Frontiers in Metallobiochemistry Symposium and Bioinorganic Workshop, State College, PA, USA. 2018.
- **Comparative genomics for the identification of unknown genes, enzymes and transporters involved in Queuosine metabolism** — Invited talk. University of Illinois at Urbana-Champaign, USA. August 2016.
- **Production of molecules of interest from plants in Yeast, can comparative genomics help?** — Invited talk. Université François Rabelais, Tours, France. May 2016.
- **Is Streptomyces secondary metabolism regulation involving lipid rafts?** — Invited talk. Université Paris 11, Orsay, France. May 2016.
- **Identification of missing enzymes and transporters involved in the synthesis and salvage of Queuosine by comparative genomics** — Poster. EMBO Conference, Ribosome Structure and Function, Strasbourg, France. 2016.
- **Identification of missing enzymes and transporters involved in the synthesis and salvage of Queuosine by comparative genomics** — Poster. DOE JGI User meeting, Genomics of Energy and Environment Meeting, Walnut Creek, CA, USA. 2016.
- **Identification of genes involved in Queuosine Salvage** — Poster. Gordon Research Conference, RNA editing, Lucca (Barga), Italy. 2015.

- **Salvage of the thiamin pyrimidine moiety by plant TenA proteins lacking an active site cysteine** — Invited talk. Plant Biology conference, Portland, OR, USA. July 2014.
- **Identification of maize and Arabidopsis genes for the salvage of thiamin breakdown products** — Poster. Gordon Research Conference, Plant Metabolic Engineering, Waterville Valley, NH, USA. 2013.
- **Identification of a potential queuine salvage gene in eukaryotes** — Poster. XXIV tRNA Conference, Olmué, Chile. 2012.
- **Identification of a triacylglycerol-lipase from Brassica napus by functional proteomics** — Poster (selected for a short oral presentation). 5th European Symposium on Plant Lipids, Gdansk, Poland. 2011.

Professional Service and Memberships

- Member, Acid Fast Club (since July 2026)
- Expert Evaluator, European Commission Horizon Europe programme
- Member, UKRI Talent Peer Review College (PRC)
- Bacterial lead, Genetically Modified Organisms Health and Safety Committee, Manchester Metropolitan University
- Carbon Literacy Champion, School of Biological and Chemical Sciences, Manchester Metropolitan University
- Member, Microbiology Society (since 2019)
- Member, NTM Network UK, Basic Science & Microbiology interest group
- Fellow of the Higher Education Academy (FHEA)